

Lending Protocols

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Why?



Alice

- * has ETH, but not USDC
- * wants USDC to pay the rent
- * doesn't want to sell ETH because she believe it will grow

- * has USDC
- * wants to earn interest



Bob

Long position

You believe ETH will rise:

- Hold ETH.
- Or put your ETH to the lending protocol and earn interest.

Short position

You believe ETH will drop:

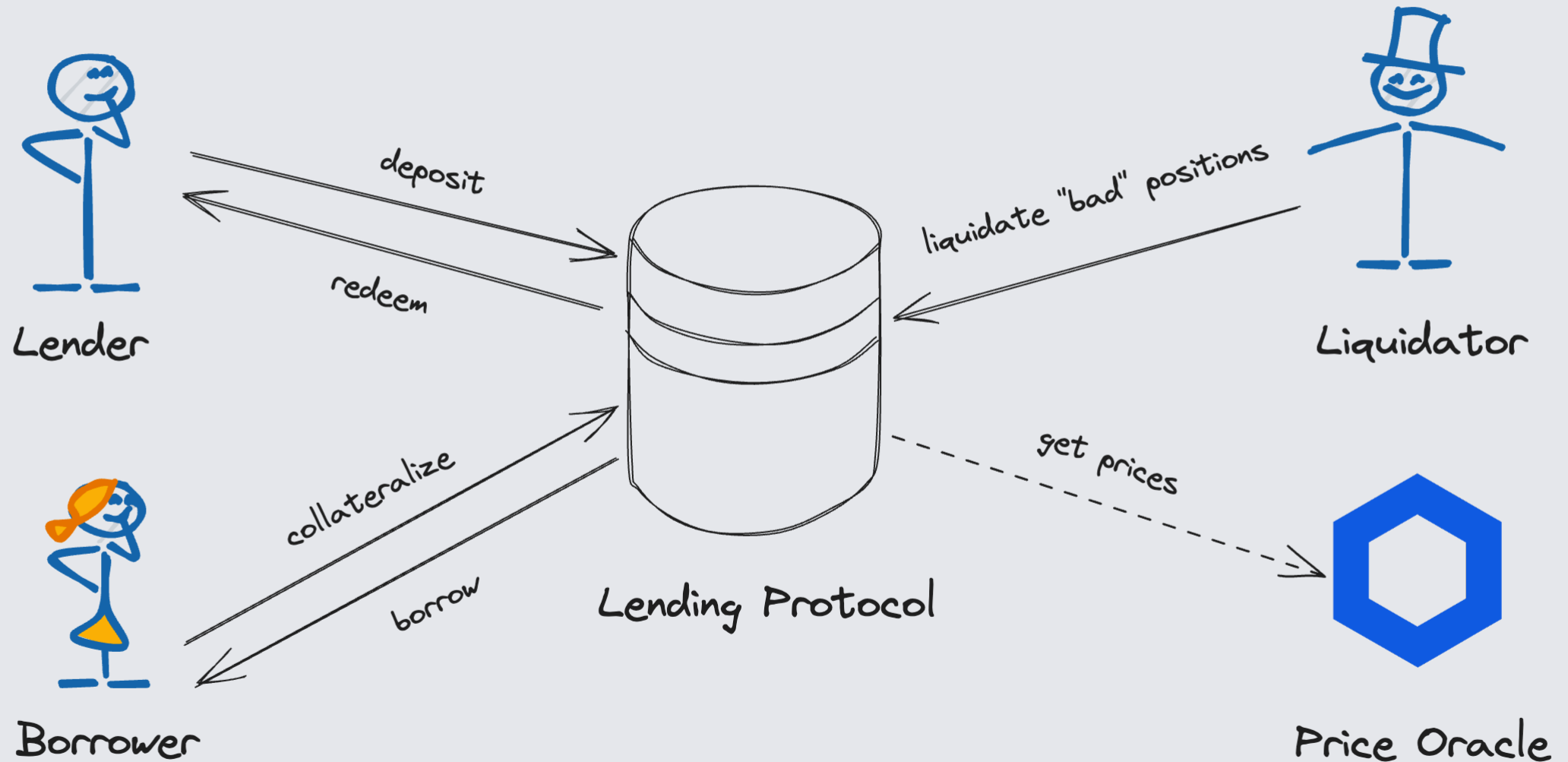
- Borrow ETH.
- Immediately sell it.
- Repay it after with a lower price.

Leveraged short position

You believe ETH will drop:

- Borrow ETH.
- Immediately sell it.
- Provide more collateral with sold ETH.
- Borrow more ETH.
- ...
- Repay it after with a lower price.

Lending actors

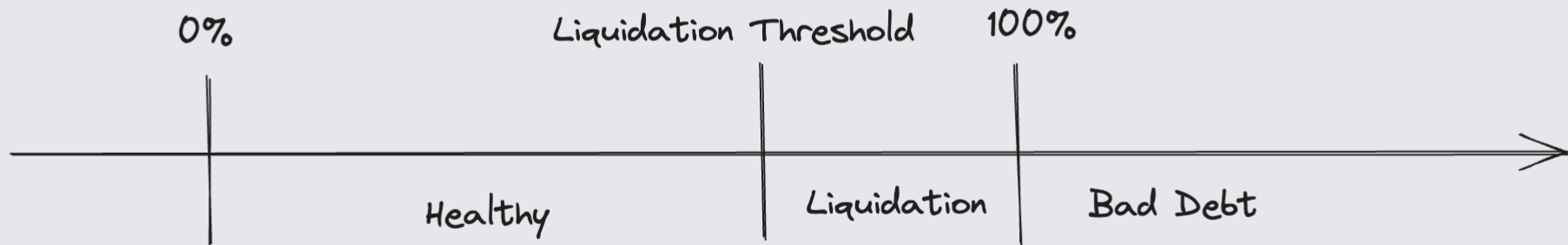


Main terms

- **Collateral:** Assets deposited by a borrower to secure a loan.
- **Liquidity Pools:** Collections of funds locked in a smart contract used for lending and borrowing.
- **Interest Rate:** The cost of borrowing or the return on lending, usually expressed as a percentage.
- **LTV (Loan-to-Value) Ratio:** The ratio of the loan amount to the value of the collateral.
- **Liquidation:** The process of selling collateral to cover a loan if its value falls below a certain threshold.

Bad debt

Debt value / Collateral value

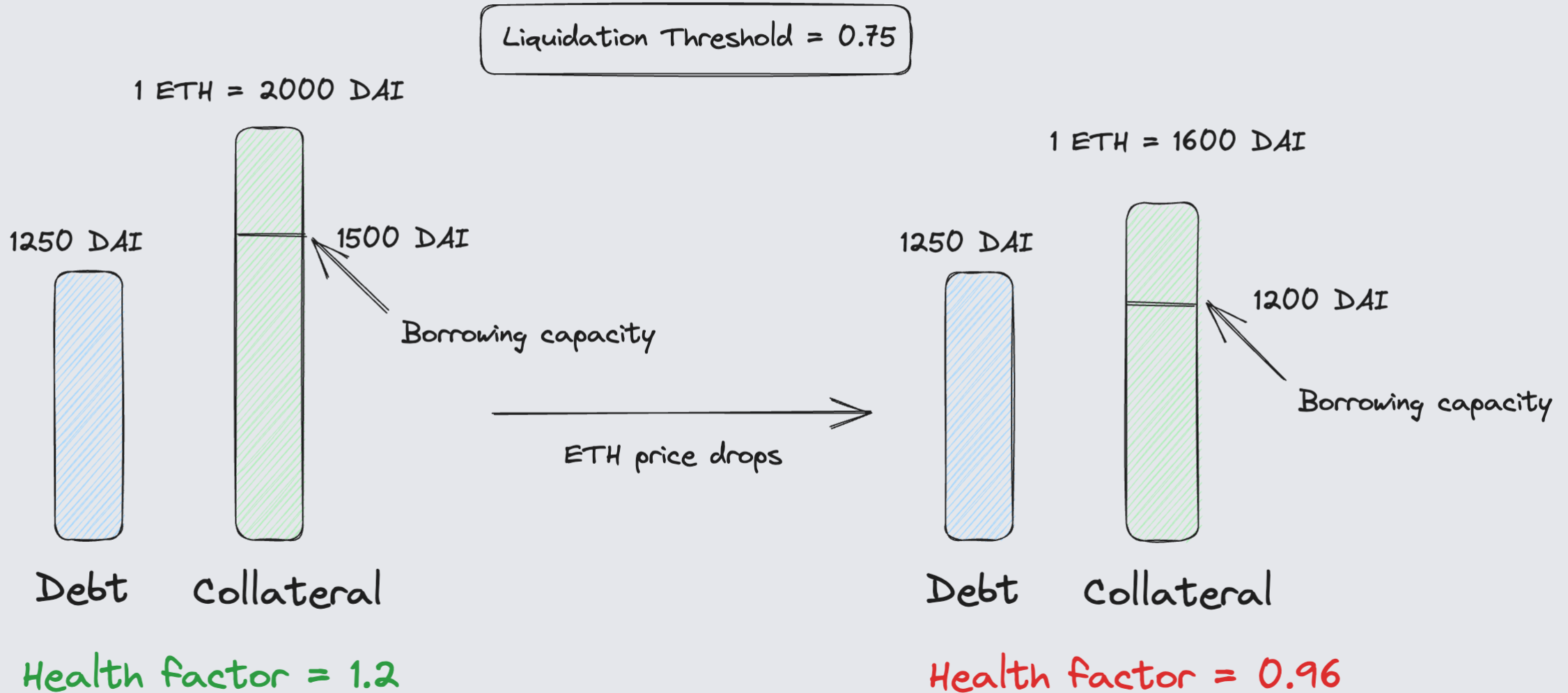


Health factor

$$\text{Health factor} = \frac{\sum \text{Value of collateral}_i \cdot \text{Liquidation threshold}_i}{\text{Total value of debts}}$$

- $0 < \text{Liquidation threshold} < 1$
- When the health factor declines below 1, a borrowing position becomes liquidatable.

Health factor



Liquidation

- Happens when health factor < 1 .
- Liquidator is repaying the user's debt and getting user's collateral.
- Liquidator needs some incentive to make it.

Liquidation terms

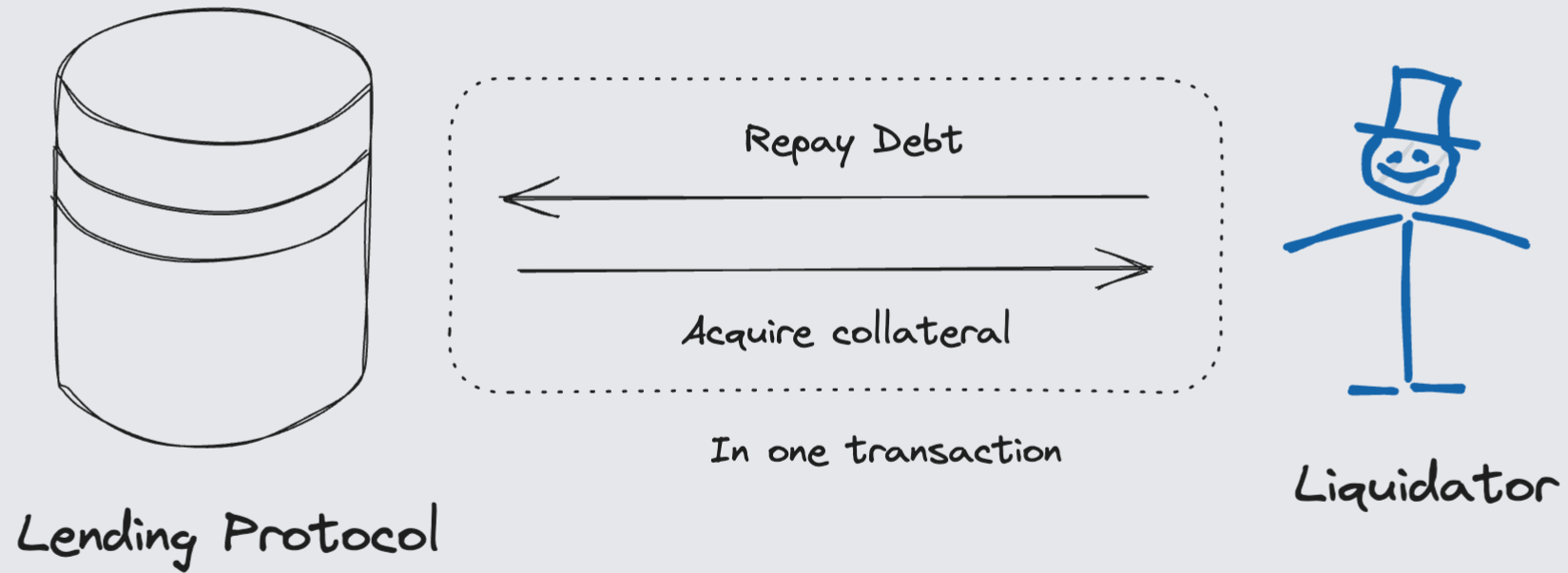
- **Liquidation Spread (LS):** Discount that liquidator gets when liquidating.

$$\textit{Value of Collateral to Claim} = \textit{Value of Debt to Repay} \cdot (1 + LS)$$

- **Close Factor (CF):** Maximum proportion of the debt that is allowed to be repaid.

$$\textit{Value of Debt to Repay} < \textit{Total Value of Debt} \cdot CF$$

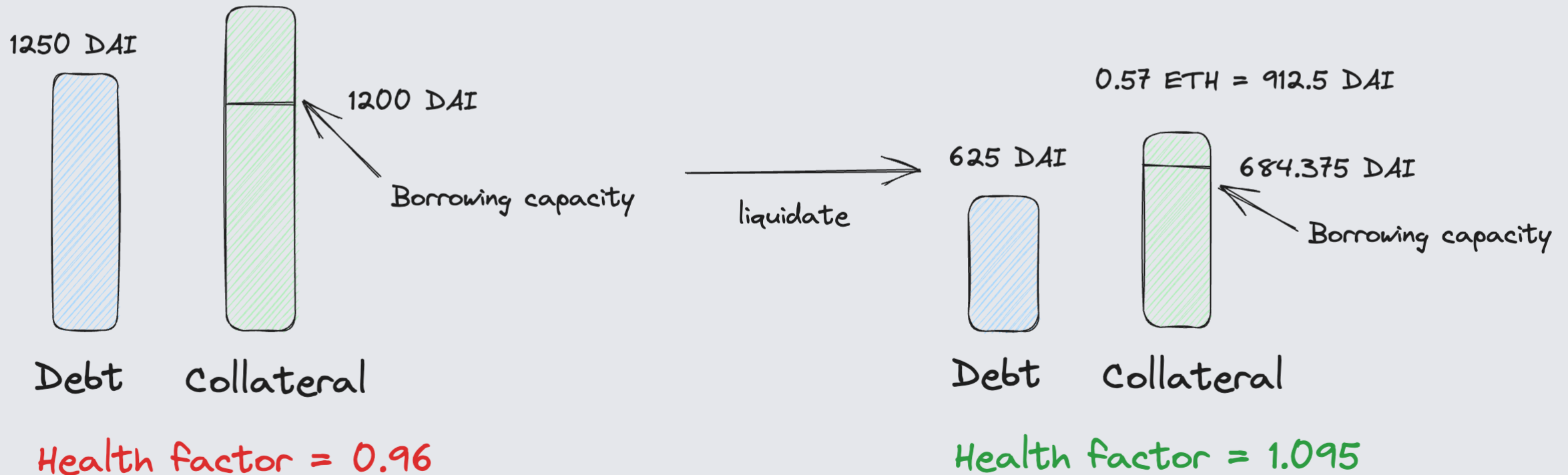
Fixed spread liquidation



Fixed spread liquidation

Liquidation Threshold = 0.75
Liquidation Spread = 0.1
Close Factor = 0.5

1 ETH = 1600 DAI

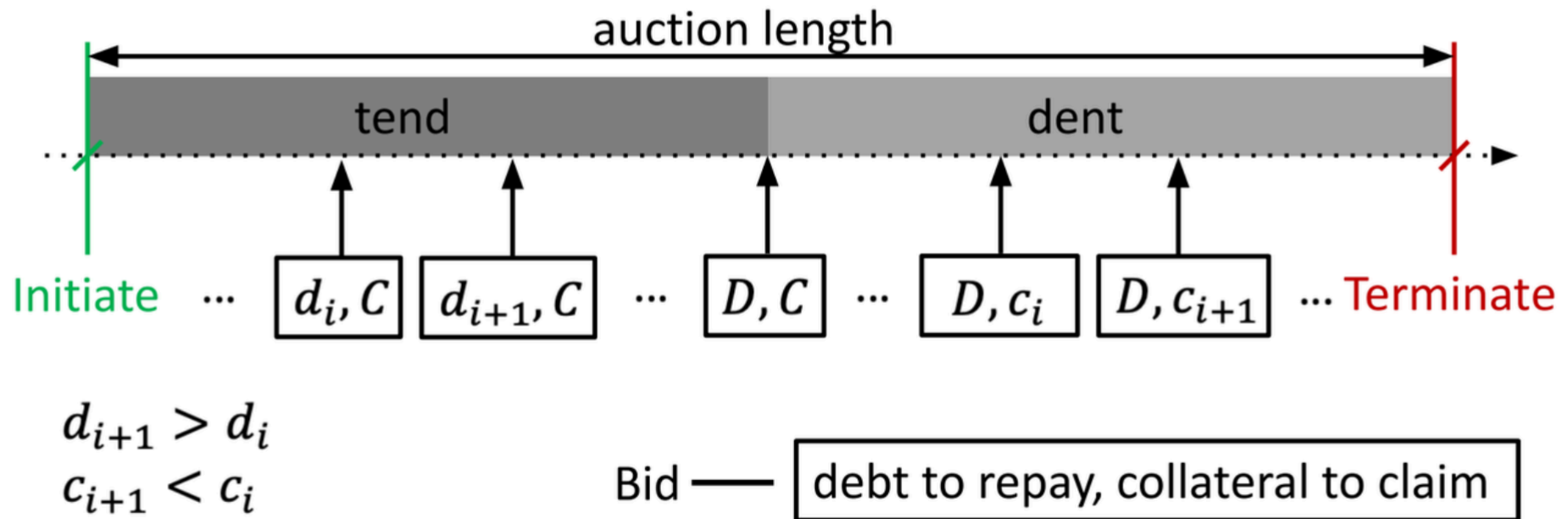


Auction liquidation

- English Auction.
 - Bidders outbid each other increasingly.
- Dutch Auction.
 - Auction begins with a high asking price and the price lowers until the auction terminates.

MakerDAO example (til April 2021)

- A position with D debt and C collateral

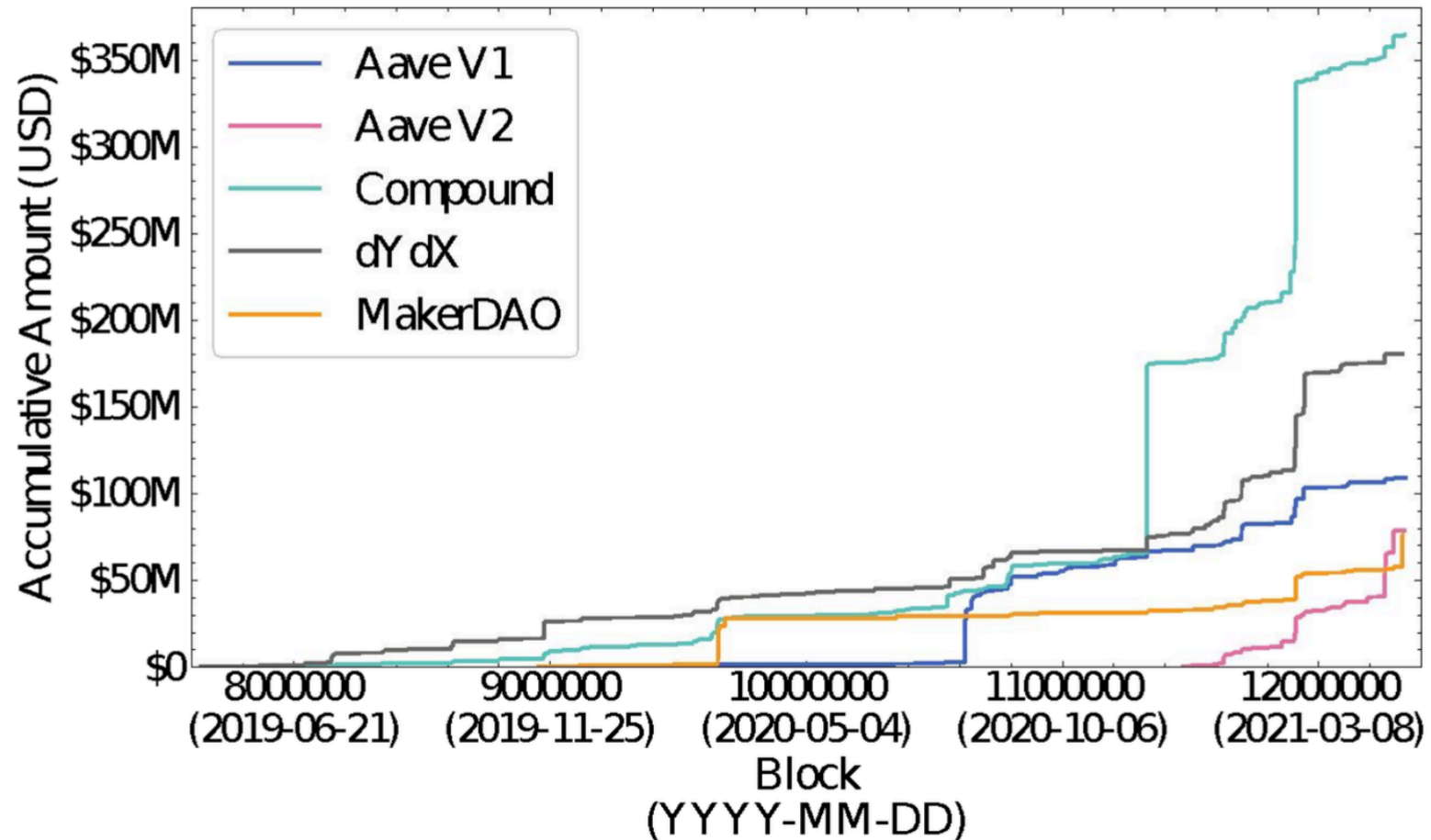


MakerDAO example (from April 2021)

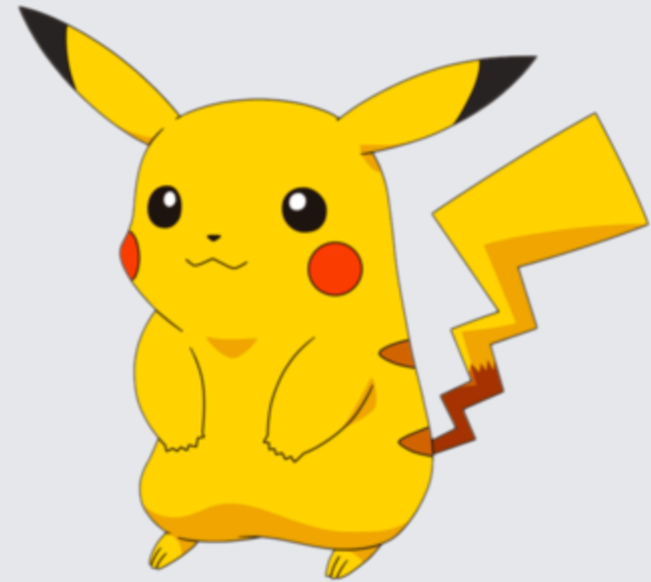
MakerDAO Dutch auction (April 2021 — Present)

- Unlike English auction which are operated in multiple transactions, the Dutch auction is settled instantly in one atomic transaction.
- No upfront debt is required => flashloan can be used.
- Collateral price decreases over time => nobody can get the collateral for free by accident.

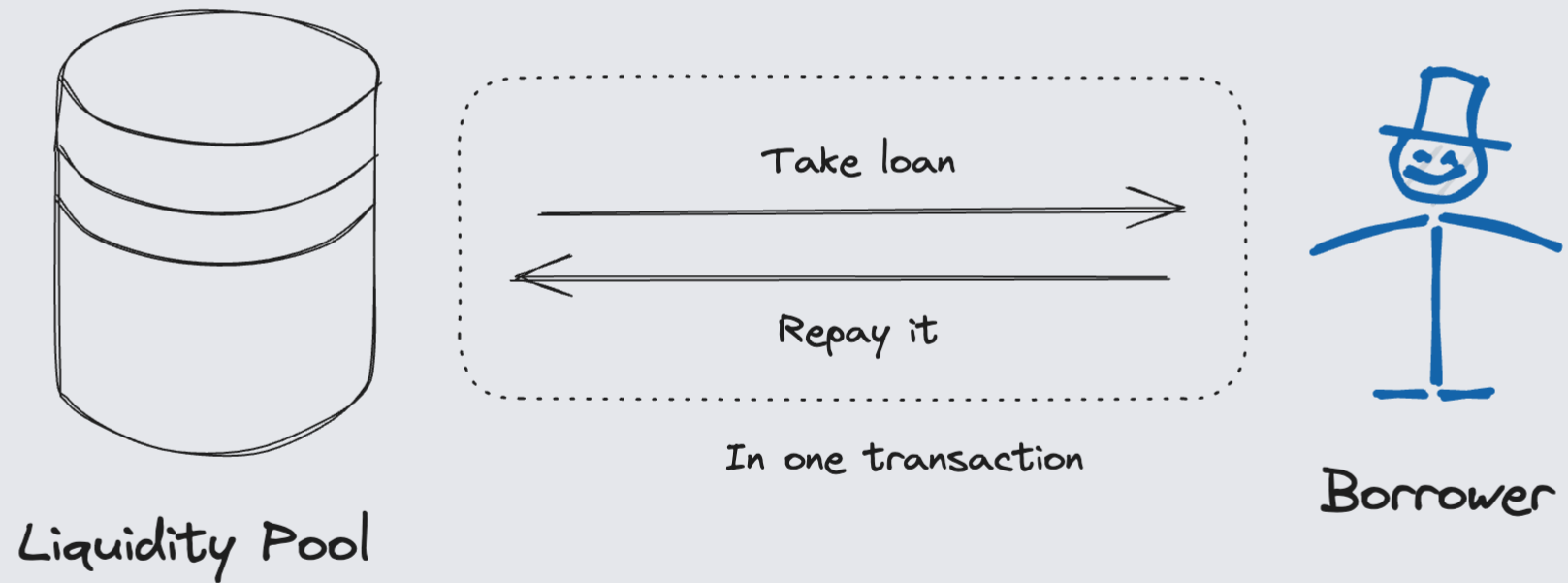
Liquidation statistics



Flashloan



Flashloan



Flashloan use cases

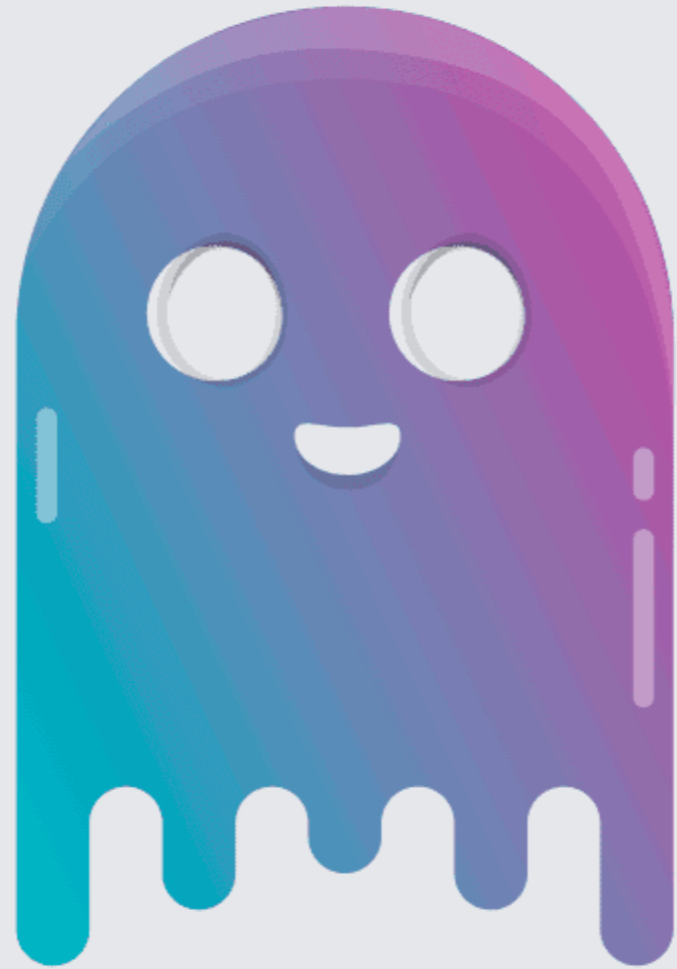
- DeFi attacks (e.g. Price Oracle manipulation).
- Arbitrage.
- Liquidations.
- etc.

Projects

Compound



Aave



MakerDAO



Curve Stablecoin

